Diagonalising the Hydrogen Hamiltonian

Liam Scarlett

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1 Theory

The radial Schrödinger equation for an electron influenced by a central potential has the form (in SI units)

$$-\frac{\hbar^2}{2m_\mathrm{e}}\frac{\mathrm{d}^2}{\mathrm{d}r^2}\psi(r)+\left[\frac{\ell(\ell+1)}{2m_\mathrm{e}r^2}+V(r)\right]\psi(r)=E\psi(r). \tag{1}$$

Here $\psi(r)$ stands for $R(r)/r$, where the three-dimensional wave function has been separated as $\Psi(r,\theta,\phi)=R(r)\Theta(\theta)\Phi(\phi)$ (in spherical coordinates).

Firstly, this equation can be simplified by rewriting it in atomic units (a.u.). In this system of units the following fundamental constants are defined to equal 1:

- electron mass m_e
- elementary charge e
- reduced Planck's constant ("h-bar") $\hbar$
- Coulomb's constant $1/(4\pi\epsilon_0)$

and some important derived units are

- length: Bohr radius $a_0\approx 5.2918\times10^{-11}\,\mathrm{m}$
- energy: hartree $E_\mathrm{h}\approx 27.2114\,\mathrm{eV}$ ($2\times$ the hydrogen ground state energy).

Then Eq. (1) becomes

$$-\frac{1}{2}\frac{\mathrm{d}^2}{\mathrm{d}r^2}\Psi(r)+\left[\frac{\ell(\ell+1)}{2r^2}+V(r)\right]\Psi(r)=E\Psi(r), \tag{2}$$

where E is measured in units of hartrees, and the potential function $V(r)$ must also be written in hartrees, for example the Coulomb potential:

$$\frac{e}{4\pi\epsilon_0 r}\ \text{(SI)}\ \rightarrow\ \frac{1}{r}\ \text{(a.u)}. \tag{3}$$

To further simplify the equation, define the Hamiltonian operator:

$$H\equiv-\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}r}+\frac{\ell(\ell+1)}{2r^2}+V(r), \tag{4}$$

which contains all the information about the kinetic and potential energies of the system. Then Eq. (2) becomes

$$H\psi(r)=E\psi(r), \tag{5}$$

which makes it clearer that we're dealing with a simple eigenvalue equation. To find a numerical solution to Eq. (5), we choose a basis of functions $\{\phi_n(r)\}_{n=1}^{\infty}$ which form a complete set. Note that that this basis

is not necessarily orthogonal. The completeness property means that any real-valued function $f(r)$ can be expressed by projecting it onto the basis, and so we write the solution to Eq. (5) as

$$\psi(r) = \sum_{m=1}^{\infty} c_m \phi_m(r), \tag{6}$$

where c_m are coefficients yet to be determined.

Substituting the expansion (6) into Eq. (5) gives

$$H \sum_{m=1}^{\infty} c_m \phi_m(r) = E \sum_{m=1}^{\infty} c_m \phi_m(r). \tag{7}$$

The Hamiltonian is a linear operator by its definition, so it can freely be moved inside the summation as follows:

$$\sum_{m=1}^{\infty} c_m H \phi_m(r) = E \sum_{m=1}^{\infty} c_m \phi_m(r). \tag{8}$$

Now multiply both sides of this equation by an arbitrary basis function $\phi_n(r)$, and integrate both sides with respect to r:

$$\int_0^{\infty} \sum_{m=1}^{\infty} c_m \phi_n(r) H \phi_m(r) \, \mathrm{d}R = \int_0^{\infty} E \sum_{m=1}^{\infty} c_m \phi_n(r) \phi_m(r) \, \mathrm{d}R \tag{9}$$

$$\sum_{m=1}^{\infty} c_m \int_0^{\infty} \phi_n(r) H \phi_m(r) \, \mathrm{d}R = E \sum_{m=1}^{\infty} c_m \int_0^{\infty} \phi_n(r) \phi_m(r) \, \mathrm{d}R \tag{10}$$

$$\sum_{m=1}^{\infty} H_{nm} c_m = E \sum_{m=1}^{\infty} B_{nm} c_m, \tag{11}$$

where we have defined the Hamiltonian matrix by its elements

$$H_{nm} = \int_0^{\infty} \phi_n(r) H \phi_m(r) \, \mathrm{d}r, \tag{12}$$

and the *overlap* matrix by its elements

$$B_{nm} = \int_0^{\infty} \phi_n(r) \phi_m(r) \, \mathrm{d}r. \tag{13}$$

Now take a closer look at the matrix equation (11). It clearly must hold for any value of n (rows in the Hamiltonian and overlap matrices), and the summation over m (columns of the Hamiltonian and overlap matrices) looks like a matrix product. In fact, for each choice of n, the left-hand side of Eq. (11) represents the nth element in the vector formed by the matrix product $H\mathbf{c}$, where $\mathbf{c}$ is the vector with elements c_m. The same also holds for the right-hand side of Eq. (11). We expect a solution for each value of n, and so we write

$$H\mathbf{c}_n = E_n B \mathbf{c}_n, \tag{14}$$

where the subscript n designates a particular eigenvalue (or eigenenergy) E_n and eigenvector $\mathbf{c}_n$. This is the standard form of a generalised eigenvalue problem, which can be solved using existing computational packages.

2 Implementation

Firstly, the numerical solutions to Eq. (14) can be obtained only when the Hamiltonian and overlap matrices are finite, i.e. the summations in Eqs. (6) and onwards must be truncated at some finite integer N. We then write

$$H^N \mathbf{c}_n^N = E_n^N B^N \mathbf{c}_n^N, \tag{15}$$

where the superscripts N remind us that the matrices have been truncated, and the eigenenergy/eigenvector solutions are not true solutions to Eq. (14), but solutions to the approximate Schrödinger equation (15). Since a complete basis has been chosen, the approximate solutions must converge to the true solutions as $N \to \infty$.

Now all that is required is to evaluate the matrix elements H_{nm} and B_{nm} for $n, m = 1, \ldots, N$, and pass them to a linear algebra subroutine (provided by LAPACK or EISPACK for example) to "diagonalise" the Hamiltonian (find the eigenvalue/vector solutions). These matrix elements can be obtained numerically by performing the integrations in Eqs. (12) and (13), but with a convenient choice of basis there will be analytical expressions for these quantities to simplify this step.

The diagonalisation subroutine will return a matrix c^N and a vector $\mathbf{E}^N$. The matrix c^N contains each eigenvector solution arranged in columns (i.e. the element c_{mn}^N contains the mth element of the eigenvector $\mathbf{c}_n^N$), and the vector $\mathbf{E}^N$ contains the eigenenergies (i.e. the element E_n^N is the energy corresponding to the eigenvector $\mathbf{c}_n^N$).

Now that the matrix eigenvalue problem has been solved, the original problem of finding the wave-function solutions $\psi(r)$ to Eq. (5) is completed by recovering the approximate solutions $\psi_n^N(r)$ using

$$\psi_n^N(r) = \sum_{m=1}^{N} c_{mn}^N \phi_m(r). \tag{16}$$

These approximate solutions are referred to a *pseudostates*, since they are not the true *states* of the hydrogen atom (true solutions to Eq. (5)). The completeness of the chosen basis, however, ensures that the pseudostates can be made to represent the true states within any required accuracy by choosing N large enough:

$$\lim_{N\to\infty} \psi_n^N(r) = \psi_n(r). \tag{17}$$

Recovering the wave functions themselves requires a numerical implementation of the basis functions, which is somewhat involved. However, a simple program can be written to generate the energies alone, requiring only the evaluation of the Hamiltonian and overlap matrix elements.

3 Matrix elements in a Laguerre basis

We use the following convenient basis constructed from Laguerre polynomials:

$$\phi_n(r) = A_{n\ell}(2\alpha r)^{\ell+1} e^{-\alpha r} L_{n-1}^{2\ell+1}(2\alpha r), \qquad n = 1, \ldots, N \tag{18}$$

where α is a parameter which can be freely chosen, $A_{n\ell}$ is a normalisation constant to be determined, and $L_{n-1}^{2\ell+1}$ is a generalised Laguerre polynomial.

In what follows, expressions are given for the overlap and Hamiltonian matrix elements. Derivations of these expressions are involved and not expected at a beginner level.

3.1 Overlap matrix and normalisation constant

The overlap matrix elements are

$$\begin{aligned} B_{nm} &= \int_0^\infty \phi_n \phi_m \, \mathrm{d}r \\ &= A_{n\ell} A_{m\ell} \int_0^\infty (2\alpha r)^{2\ell+2} e^{-2\alpha r} L_{n-1}^{2\ell+1}(2\alpha r) L_{m-1}^{2\ell+1}(2\alpha r) \, \mathrm{d}r. \end{aligned} \tag{19}$$

We evaluate the above using the orthogonality of the generalised Laguerre polynomials, obtaining

$$\begin{aligned} B_{nm} = \frac{1}{\alpha} A_{n\ell}^2 \frac{(n+\ell)(n+2\ell)!}{(n-1)!} \delta_{nm} - \frac{1}{2\alpha} A_{n\ell} A_{(n+1)\ell} \frac{(n+2\ell+1)!}{(n-1)!} \delta_{n,m-1} \\ - \frac{1}{2\alpha} A_{n\ell} A_{(n-1)\ell} \frac{(n+2\ell)!}{(n-2)!} \delta_{n-1,m}, \end{aligned} \tag{20}$$

where δ_{nm} is the Kronecker delta function. We now choose the normalisation constant $A_{n\ell}$ in such a way as to yield unity along the main diagonal of the overlap matrix:

$$A_{n\ell} \equiv \sqrt{\frac{\alpha(n-1)!}{(n+\ell)(n+2\ell)!}}, \tag{21}$$

giving

$$B_{nm} = \delta_{nm} - \frac{1}{2}\sqrt{1 - \frac{\ell(\ell+1)}{(n+\ell)(n+\ell+1)}} \delta_{n,m-1} - \frac{1}{2}\sqrt{1 - \frac{\ell(\ell+1)}{(n+\ell-1)(n+\ell)}} \delta_{n-1,m}. \tag{22}$$

This is the form of a tridiagonal matrix, with the three terms describing the diagonal, superdiagonal, and subdiagonal:

$$\delta_{nm} : \text{diagonal—row and column indices equal } (n = m) \tag{23}$$

$$\delta_{n,m-1} : \text{superdiagonal—column index one greater than row index } (m = n + 1) \tag{24}$$

$$\delta_{n-1,m} : \text{subdiagonal—column index one less than row index } (m = n - 1). \tag{25}$$

The subdiagonal term in Eq. (22) can be obtained from the superdiagonal term with the replacements $n \to n - 1$ and $m \to m + 1$, which describes a reflection about the diagonal. Hence, the overlap matrix is symmetric, as we should expect by inspection of Eq. (19).

The overlap matrix can be constructed by setting the diagonal elements equal to 1, setting the superdiagonal elements equal to

$$-\frac{1}{2}\sqrt{1 - \frac{\ell(\ell+1)}{(n+\ell)(n+\ell+1)}}, \tag{26}$$

and then reflecting the superdiagonal about the diagonal onto the subdiagonal. All other elements of the matrix (above the superdiagonal and below the subdiagonal) should be set to zero.

3.2 Hamiltonian matrix

The kinetic-energy operator (the Hamiltonian minus the potential term) is

$$K \equiv -\frac{1}{2}\frac{\mathrm{d}^2}{\mathrm{d}r^2} + \frac{\ell(\ell+1)}{2r^2}. \tag{27}$$

Hence, the Hamiltonian matrix elements are given by

$$\begin{aligned} H_{nm} &= \int_0^\infty \phi_n(r)\left[K + V(r)\right]\phi_m(r)\,\mathrm{d}r \\ &= \int_0^\infty \phi_n(r) K \phi_m(r)\,\mathrm{d}r + \int_0^\infty \phi_n(r) V(r) \phi_m(r)\,\mathrm{d}r \\ &= K_{nm} + V_{nm} \end{aligned} \tag{28}$$

The matrix elements of K and V (for hydrogen) are

$$K_{nm} = -\frac{\alpha^2}{2} B_{nm} + \alpha^2 \delta_{nm} \tag{29}$$

$$V_{nm} = -\frac{\alpha}{(n+\ell)} \delta_{nm}. \tag{30}$$

Note that K can be expressed in terms of B, so it is best to first evaluate B and then use it in evaluating K. The simple analytical form of V is possible only for hydrogen-like atoms. In general, for an atom with a single electron and a nuclear charge of Z, the V-matrix elements are

$$V_{nm} = -\frac{Z\alpha}{(n+\ell)} \delta_{nm}. \tag{31}$$

For any other system, the matrix elements must be obtained by numerically evaluating

$$V_{nm} = \int_0^\infty \phi_n(r) V(r) \phi_m(r)\,\mathrm{d}r. \tag{32}$$